

Company: Alvin Tech

Security Scan Report (Post-Patching)

Report Date: March 25, 2025

Prepared by: InfoSec Department

Scan Period: March 10-20, 2025

Executive Summary

This report presents the findings of comprehensive security scans conducted after the completion of our major security patching initiative. The scans were performed using multiple industry-standard tools to evaluate the effectiveness of our remediation efforts in addressing previously identified vulnerabilities.

Methodology

- External vulnerability scanning using Tenable Nessus Pro
- Web application scanning using OWASP ZAP
- Network penetration testing using Metasploit Framework
- Database security assessment using Database Security Scanner Pro
- PCI DSS compliance scanning using Approved Scanning Vendor (ASV) tools

Key Findings

1. Vulnerability Statistics

Severity	Previous Count	Current Count	Reduction
Critical	28	3	89%
High	64	12	81%
Medium	147	35	76%
Low	215	89	59%
Informational	283	196	31%

2. External Vulnerability Scan

System Group	Critical Findings	High Findings	Medium Findings	Notes
Public API Endpoints	0	1	4	High: TLS configuration weakness
Website Infrastructure	0	2	7	High: Outdated web server components
VPN Access Points	0	0	3	Medium: Certificate validation issues
Email Infrastructure	1	2	4	Critical: Mail server vulnerability CVE-2024-8976

3. Internal Infrastructure Scan

System Group	Critical Findings	High Findings	Medium Findings	Notes
Payment Processing Servers	0	0	3	Acceptable risk level
Database Servers	1	2	5	Critical: Privilege escalation vulnerability in PostgreSQL
Application Servers	0	2	6	High: Two outdated libraries
Admin Workstations	1	3	8	Various endpoint protection issues

4. Web Application Scan

Application	Critical Findings	High Findings	Medium Findings	Notes
Customer Portal	0	1	3	High: Potential XSS vulnerability
Admin Dashboard	0	0	2	Improved significantly
Payment Gateway	0	1	1	High: CSRF vulnerability
API Services	0	0	2	API rate limiting issues

5. PCI DSS Compliance Scan

The compliance scan was conducted using an Approved Scanning Vendor (ASV) tool.

Requirement	Compliant	Notes
Req 1: Network Security	Partial	Firewall rule inconsistencies detected
Req 2: Secure Configurations	Partial	Default credentials found on 2 non-critical systems
Req 3: Cardholder Data Protection	Yes	Encryption standards meet requirements
Req 4: Transmission Protection	Yes	All transmissions use TLS 1.2+
Req 5: Malware Protection	Yes	Anti-malware solutions appropriately deployed
Req 6: Secure Systems/Applications	Partial	3 systems have outdated components
Req 8: Access Control	Partial	Weak password policies for some internal systems
Req 10: Monitoring	Yes	Logging and monitoring systems adequate
Req 11: Testing	Yes	Regular testing is occurring

Remediation Recommendations

Critical Issues (Immediate Action Required)

- 1. Mail server vulnerability (CVE-2024-8976)**
 - Apply vendor patch immediately
 - Estimated time: 1 day
- 2. PostgreSQL privilege escalation vulnerability**
 - Apply database patch and review access controls
 - Estimated time: 2 days
- 3. Admin workstation security**
 - Deploy endpoint protection updates
 - Update security policies
 - Estimated time: 3-5 days

High Priority Issues (Action within 2 weeks)

1. TLS configuration weaknesses
2. Outdated web server components

3. XSS and CSRF vulnerabilities
4. Remaining outdated libraries

Medium Priority Issues (Action within 1 month)

1. PCI DSS compliance gaps
2. Certificate validation issues
3. API rate limiting
4. Remaining medium-severity vulnerabilities

Conclusion

While significant progress has been made in addressing the previously identified security vulnerabilities, several critical and high-priority issues remain. These issues require immediate attention to achieve full compliance with PCI DSS and secure the company's virtual card and wallet infrastructure.

The scan results indicate approximately 85% of previously identified issues have been successfully remediated. A follow-up scan is recommended after addressing the remaining critical and high-priority issues.

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